Body Control Module - High-Level Design

Document version 1.1

1 Introduction

1.1 Purpose

The purpose of this document is to describe the high-level software architecture of the Body Control Module (BCM), including its software components, ports, interfaces, signals and dependencies.

1.2 Scope

This document covers the application layer components of the BCM. Basic software configuration is described in separate documents.

2 System Overview

2.1 Architecture Overview

The BCM application layer consists of the following software components: DoorLockManager, WindowController, LightingController, PowerManager, DiagnosticHandler, ComSignalRouter, RainSensorHandler. Components communicate through AUTOSAR ports typed by interfaces, and exchange vehicle signals through the ComSignalRouter.

2.2 Component Catalogue

Component	Description
DoorLockManager	Locks and unlocks the vehicle doors based on vehicle speed and remote requests.
WindowController	Controls power window movement and anti-pinch protection.
LightingController	Manages exterior and interior lighting based on light requests.
PowerManager	Distributes the vehicle power state to all body components.
DiagnosticHandler	Handles UDS diagnostic requests and routes them to body components.
ComSignalRouter	Routes signals between the body components and the vehicle network.
RainSensorHandler	Reads the rain sensor and publishes the rain intensity.

3 Software Architecture

3.1 Interfaces

The following interfaces are defined for the BCM application layer.

Interface	Description
IDoorLock	Lock and unlock commands for the door actuators.
IWindowCtrl	Window movement commands.
ILightCtrl	Light switching commands.
IPowerState	Current vehicle power state notifications.
IDiagService	Diagnostic service request interface.
ISignalRoute	Signal routing interface towards the network stack.
IRainStatus	Rain intensity notifications.

3.2 Ports

Each component exposes provided and required ports. Required ports must be connected to a matching provided port.

Component	Port	Direction	Interface
DoorLockManager	DoorLockCmd	provided	IDoorLock
DoorLockManager	PowerStatePort	required	IPowerState
WindowController	WindowCmd	provided	IWindowCtrl
LightingController	LightCmd	provided	ILightCtrl
LightingController	PowerStatePort	required	IPowerState
PowerManager	PowerStateOut	provided	IPowerState
DiagnosticHandler	DiagReq	provided	IDiagService
DiagnosticHandler	DoorLockDiag	required	IDoorLockDiag
ComSignalRouter	SignalOut	provided	ISignalRoute
RainSensorHandler	RainStatusOut	provided	IRainStatus

3.2.1 Power State Handling

The PowerManager provides the power state through the IPowerState interface. DoorLockManager, LightingController and WindowController require the power state to enter low-power mode safely.

4 Communication

4.1 Signal List

Signal	Sender	Receiver	Cycle Time
DoorLockStatus	DoorLockManager	ComSignalRouter	100 ms
WindowPosition	WindowController	ComSignalRouter	100 ms
VehSpeed	ComSignalRouter	DoorLockManager	20 ms
IgnitionState	PowerManager	ComSignalRouter	50 ms
LightRequest	ComSignalRouter	LightingController	50 ms
DiagSessionState	DiagnosticHandler	-	event
RainIntensity	RainSensorHandler	WindowController	50 ms

4.2 Component Dependencies

Component	Depends On	Type
DoorLockManager	PowerManager	service
WindowController	PowerManager	service
LightingController	PowerManager	service
DoorLockManager	ComSignalRouter	data
DiagnosticHandler	DoorLockManager	diagnostic
LightingController	AmbientSensorDriver	data
WindowController	RainSensorHandler	data

5 Diagnostics

The DiagnosticHandler receives UDS requests over the IDiagService interface. Requests addressing door locks are forwarded to the DoorLockManager through the DoorLockDiag port. The current diagnostic session is published as the DiagSessionState signal.

6 Revision History

Version	Date	Change
1.0	2026-01-10	Initial release
1.1	2026-03-02	Added RainSensorHandler; renamed VehicleSpeed; updated ports.